

KHADAMA, Russian Federation

WMO: 298920

Lat: 53.950N

Lon: 98.817E

Elev: 2313

StdP: 13.51

Time Zone: 8.00 (E08)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-28.6	-37.8	0.7	-33.6	-32.7	0.9	-28.2	12.2	8.2	8.5	10.1	1.9	170	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.1	81.6	61.8	77.2	60.3	73.5	59.0	64.7	75.6	62.8	72.8	60.9	69.9	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.3	88.3	68.0	59.3	82.2	65.8	57.5	76.9	63.7	31.1	75.2	29.6	73.3	28.2	69.8	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	6.7	5.3	-39.3	88.8	6.0	2.2	-43.6	90.4	-47.1	91.7	-50.5	93.0	-54.8	94.6		
(5)			-39.3	69.5	5.7	1.8	-43.4	70.7	-46.8	71.8	-50.0	72.8	-54.2	74.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	-7.0	1.9	16.1	31.9	44.7	53.8	60.4	56.8	44.5	31.6	11.6	-4.6	(6)	
	DBStd	24.94	10.80	10.16	9.99	8.06	7.75	6.48	5.49	5.17	6.67	8.79	11.88	11.96	(7)	
	HDD50	8557	1767	1346	1050	543	202	34	4	8	190	570	1152	1691	(8)	
	HDD65	13294	2232	1766	1515	993	628	338	159	256	614	1035	1602	2156	(9)	
	CDD50	758	0	0	0	0	39	148	326	220	25	0	0	0	(10)	
	CDD65	21	0	0	0	0	0	3	15	3	0	0	0	0	(11)	
	CDH74	725	0	0	0	0	82	174	355	103	11	0	0	0	(12)	
	CDH80	159	0	0	0	0	14	44	82	18	1	0	0	0	(13)	
	WSAvg	1.9	1.8	1.8	2.0	2.4	2.3	1.6	1.6	1.6	1.6	2.0	2.0	1.7	(14)	
	PrecAvg	20.7	0.2	0.2	0.4	0.9	2.1	3.5	4.7	4.5	2.5	0.7	0.4	0.3	(15)	
(16) Precipitation	PrecMax	26.6	0.6	0.3	0.7	1.6	3.3	5.3	7.8	7.4	5.4	1.1	1.1	0.6	(16)	
	PrecMin	17.4	0.0	0.0	0.0	0.4	1.1	1.7	2.6	2.0	1.4	0.2	0.2	0.0	(17)	
	PrecStd	2.4	0.1	0.1	0.2	0.3	0.7	0.9	1.1	1.5	1.0	0.3	0.2	0.2	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	34.4	53.3	65.0	82.3	85.2	86.4	82.1	75.2	62.7	43.7	30.7	(19)	
		MCWB	21.7	26.6	40.4	47.4	57.7	61.7	64.7	63.6	57.8	47.7	34.3	25.2	(20)	
	2%	DB	19.1	27.8	45.6	57.6	75.4	79.1	82.3	76.4	67.2	55.5	36.3	22.7	(21)	
		MCWB	16.7	22.2	35.3	42.9	54.2	59.2	63.7	61.3	53.9	42.9	30.4	19.6	(22)	
	5%	DB	14.0	23.8	39.4	52.1	68.7	74.1	78.1	72.4	61.0	48.9	31.3	17.4	(23)	
		MCWB	12.1	19.7	31.4	39.9	50.9	56.9	62.3	59.7	50.4	39.6	27.1	16.0	(24)	
	10%	DB	9.4	19.1	34.5	47.0	63.1	69.5	74.0	68.2	55.9	43.9	27.7	12.0	(25)	
		MCWB	8.0	15.9	28.1	36.9	48.1	55.3	61.4	58.5	48.0	36.7	24.7	11.0	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.1	27.3	40.3	47.8	58.5	65.1	68.1	65.4	59.3	48.0	35.5	26.1	(27)	
		MCDB	24.5	33.8	52.7	64.0	76.9	78.9	79.3	74.6	71.1	61.9	43.2	29.9	(28)	
	2%	WB	16.7	22.7	35.3	43.2	55.5	61.2	65.8	63.0	54.8	43.7	30.9	19.6	(29)	
		MCDB	19.0	27.2	44.7	56.7	73.3	73.2	77.0	72.1	65.2	54.0	36.0	22.0	(30)	
	5%	WB	12.1	19.8	31.6	40.0	52.2	59.1	64.0	61.1	51.9	40.5	27.5	15.8	(31)	
		MCDB	13.8	23.8	38.7	51.3	65.8	69.8	74.5	70.0	59.4	47.4	30.8	17.5	(32)	
	10%	WB	8.0	16.1	28.3	37.2	49.1	56.9	62.4	59.3	49.1	37.5	24.8	11.2	(33)	
		MCDB	9.4	19.3	34.3	45.7	61.3	66.8	71.9	66.7	54.8	42.7	27.7	12.1	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	21.2	31.1	33.5	25.0	26.1	26.1	24.1	21.9	22.3	23.0	19.6	18.1	(35)	
		MCDBR	24.0	35.5	38.8	35.5	40.4	37.4	33.9	31.2	33.1	32.3	21.7	20.8	(36)	
		MCWBR	21.3	29.6	28.4	21.7	20.8	19.1	16.8	16.4	19.3	20.0	16.3	18.0	(37)	
	5% WB	MCDBR	23.3	35.1	37.2	33.0	35.6	30.6	28.7	27.4	31.5	29.8	20.8	20.7	(38)	
		MCWBR	20.9	29.4	26.9	20.2	19.4	16.9	15.5	15.9	19.3	19.0	16.1	17.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.257	0.289	0.328	0.365	0.389	0.393	0.396	0.373	0.332	0.309	0.283	0.258	(40)	
	taud		2.162	2.148	2.146	2.230	2.245	2.343	2.370	2.408	2.468	2.405	2.216	2.110	(41)	
	Ebn at Noon		222	252	267	272	272	272	268	268	268	247	208	191	(42)	
	Edh at Noon		20	29	36	39	41	37	36	32	27	23	19	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		325	624	1073	1471	1620	1758	1597	1385	1002	606	325	222	(44)	
	RadStd		21	39	50	73	138	113	137	110	94	45	30	18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page